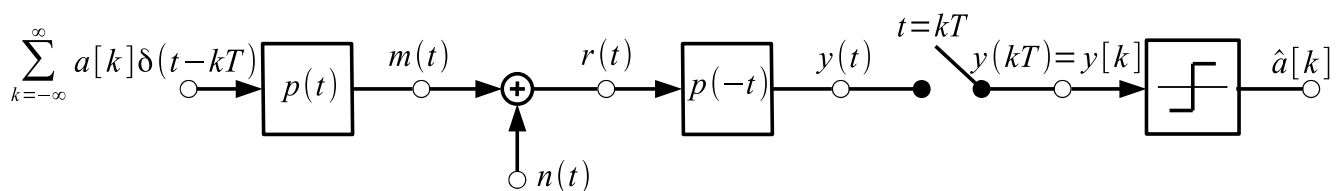


Lab-Bipolar-NRZ

Bipolar NRZ

The following block diagram shows a binary **baseband communication system** using **bipolar** symmetric **non-return-to-zero NRZ** line codes



with the following discrete- and continuous-time signals:

Transmitted bipolar binary symbol sequence: $a[k] \in \{\pm 1\}$

Transmit impulse: $p(t)$

Modulated transmit signal: $m(t)$

Additive white Gaussian noise: $n(t)$

Matched filter impulse response: $h(t) = p(-t)$

Matched filter output signal: $y(t)$

Matched filter output signal sampled at **symbol rate**
 $r = 1/T$: $y[k] = y(kT)$, where T is the **symbol period**

Note that k is a discrete time index at the symbol rate or simply the **symbol index**.

This means that $m(t)$ is the transmitted signal, i.e., the output signal of the transmitter and $r(t)$ is the received signal, i.e., the input signal of the receiver. From the transmitter, the transmitted signal propagates to the receiver over the communication channel. In the above example additive white Gaussian noise (**AWGN**) $n(t)$ is the only channel impairment and the channel is thus referred

to as an **AWGN channel**.

For the generation of the modulated transmit signal $m(t)$, the transmit symbols $a[k]$ first become weights of an impulse train $\delta(t - kT)$ of Dirac impulses spaced by the symbol duration T :

$$\sum_{k=-\infty}^{\infty} a[k]\delta(t - kT),$$

which is then input to the transmit filter with impulse response $p(t)$ to yield the modulated signal $m(t)$:

$$m(t) = \left(\sum_{k=-\infty}^{\infty} a[k]\delta(t - kT) \right) * p(t) = \sum_{k=-\infty}^{\infty} a[k]\delta(t - kT) * p(t) = \sum_{k=-\infty}^{\infty} a[k]p(t - kT),$$

The last identity of the above equation uses the fact that the convolution of the transmit pulse $p(t)$ with a shifted Dirac impulse $\delta(t - kT)$ results in a time shifted transmit pulse: $p(t) * \delta(t - kT) = p(t - kT)$.

Thus, for sequentially transmitting the binary information symbols $a[k]$ at a symbols rate $r = 1/T$, the transmit pulse $p(t)$ is transmitted every T seconds with **amplitude shift keying (ASK)**, where the k -th information symbol $a[k]$ becomes the amplitude of the transmit pulse $p(t - kT)$ transmitted at the k -th symbol period.

In the GNUradio simulation, the analog signals $p(t)$, $m(t)$, $h(t)$ will be replaced by discrete-time signals $p[n]$, $m[n]$, $h[n]$ of a certain **sampling rate $r_s = 1/T_s$** . Usually the **sampling rate r_s** is an integer multiple of the **symbol rate r** :

$$r_s = N \cdot r.$$

Then N denotes the **number of samples per symbol (SPS)** and the modulated discrete-time transmit signal can be described as

$$m[n] = \underbrace{\left(\sum_{k=-\infty}^{\infty} a[k] \delta[k - kN] \right)}_{a_{up}[n]} * p[n] = a_{up}[n] * p[n] = \sum_{k=-\infty}^{\infty} a[k] \delta[k - kN] * p[n] =$$

The input signal $a_{up}[n]$ of the transmit filter with impulse response $p[n]$ is simply an **upsampled version** of $a[k]$, obtained by inserting **$N-1$** zeros between every symbol $a[k]$:

$$a_{up}[n] = \begin{cases} a[n/N] & , \text{ if } n/N \text{ is integer} \\ 0 & , \text{ otherwise} \end{cases}$$

Thus, there are **no Dirac-impulses in the discrete-time implementation.** The upsampling factor N can also be seen as the discrete-time symbol period, i.e., the discrete-time counterpart of the symbol duration $T = N \cdot T_s$, where T_s is the **sampling period**.

In the receiver, a corresponding downsampling operation from sample to symbol rate is performed as a decimation by a factor of N after the receive filter, yielding the received signal $y[k]$ at symbol rate:

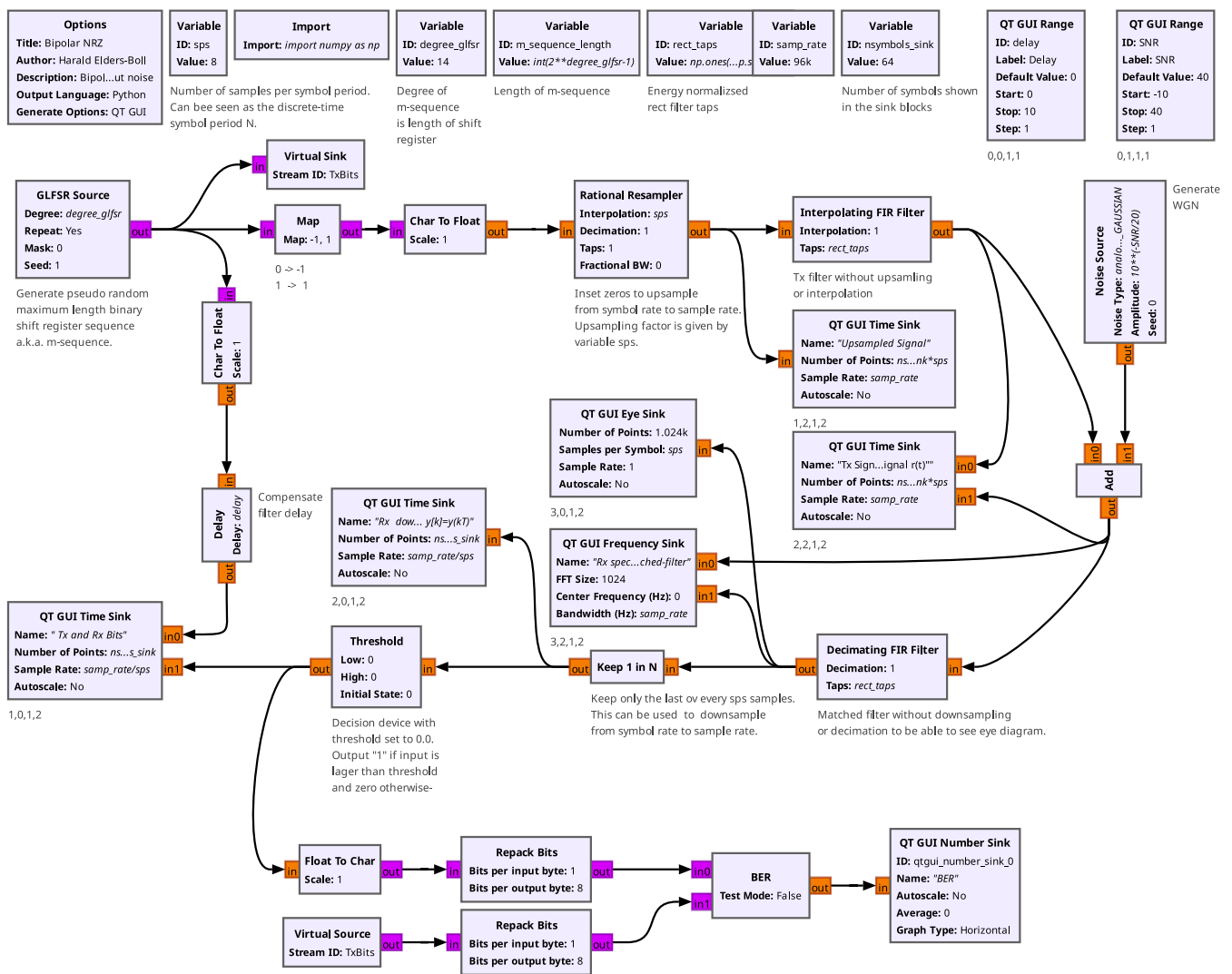
$$y[k] = y[n = k \cdot N].$$

In case of transmission over the AWGN channel, it can be shown that receive filter, whose impulse response $h[n]$ is a

time-reversed version of the transmit pulse, i.e., $h[n] = p[-n]$, is the optimum receive filter with respect to a maximization of the **Signal-to-noise ratio (SNR)**. By having an impulse response that is a time-reversed version of the transmit filter, the receive filter is in this sense matched to the transmit pulse and therefore called the **matched filter**.

GNUradio Simulation with Rectangular Pulse Shape

Now start the GNUradio companion (gnuradio-companion) and open the [GNUradio](#) flowgraph BipolarNRZrect.grc :



To be able to see the configuration of the parameters, especially the Sample Rate parameter of the different blocks, select Show parameter expressions in block and deselect Show parameter value in block in the View menu of the GNUradio companion.

Examine the flowgraph carefully and read the comments below the blocks to understand it well enough!

Run the flowgraph and identify the different signals that are being plotted in the window that opens up.

Answer the following questions and note the answers such that you have them available when presenting the lab results in the office hours:

What is the name of the samples per symbol parameter N in this flowgraph? What value is it set to?

What is the sampling rate r_s used in this flowgraph?

How are the bits $b[k] \in \{0,1\}$ mapped to the symbols $a[k] \in \{\pm 1\}$?

Note that QT GUI Time Sink blocks plot signals as line plots generating an impression that the signal were analog while in fact they are being discrete-time. In this flowgraph, line plots are used for the analog signals in the block diagram above, i.e., for $m(t)$ and $y(t)$ or the corresponding discrete-time signals at sampling rate, whereas a stem plot is used for the upsampled symbol sequence $a_{up}[n]$.

In this example, a rectangular pulse shape is used for both the impulse response $p[n]$ of the transmit filter and the impulse response $h[n]$ the received filter, with N non-zero values that are all equal to $1/\sqrt{N}$. The value of $1/\sqrt{N}$ is used to normalize the energy of the impulse response to

one: $E_p = \sum_{-\infty}^{\infty} |p[n]|^2 = 1.$

First, the **Signal-to-noise ratio (SNR)** is set to a relatively high value of 40dB such that there is hardly any noise. This can be seen from the time domain plot of the transmitted (Tx) signal $m(t)$ and the received (Rx) signal $r(t)$, because there is almost no visible difference between the two. Note that you can deselect or hide a plot by clicking on the corresponding legend on the right edge of the corresponding plot window. If the red curve is shown, deselect $r(t)$ to actually see the blue curve, or vice versa. Click the corresponding legend a second time to show the hidden curve.

Therefore the transmitted (Tx) bits and the received (Rx) bits as shown in the corresponding plot are all equal. Again hide one curve to see the other and then unhide it again.

Now examine the **eye diagram** of the received signal $y(t)$ after the matched filter.

The **eye diagram** can be used to check whether there is **intersymbol interference (ISI)** or not and how much noise is still present after the matched filter. For the eye diagram, multiple segments of the output signal of the receive filter are plotted periodically to the same figure window.

When there is no ISI and no noise, the eye diagram possesses a maximum eye-opening at multiples of the symbol period, that is steady, i.e., not changing over time.

Without additional gains, the value at these positions always corresponds to either E_p , when the corresponding symbol $a[k]$ transmitted at symbol index k was equal to one: $a[k] = 1$, or to $-E_p$, when $a[k] = -1$ was transmitted, without

any interference from previous or future symbols.

Note that with the rectangular pulse shape, these points are located at a sample counter that corresponds to $N-1$ (and $2N-1$) in the eye diagram, since the sample counter start at zero.

Only the samples at these positions are the remaining symbols after downsampling the matched filter output signal $y(t)$ (or $y[n]$ in GNUradio) from sample to symbol rate: $y[k] = y(t = nT)$ or $y[k] = y[n = kN]$. Without additional gains, without ISI and, at an SNR of 40 dB, practically without noise, these samples are all equal to plus or minus E_p : $y[k] = \pm E_p$, which yields $y[k] = \pm E_p = \pm 1$ in this case, due to the energy-normalization of $p[n]$.

These “soft” symbols $y[k]$ are input to the decision device that decides, which symbol has been sent, by comparing the soft symbols $y[k]$ against a threshold to yield “hard” symbols decisions. In the case of bipolar transmission, that is equal to zero:

$$\hat{b}[k] = \begin{cases} 1 & , \text{ if } y[k] \geq 0 \\ 0 & , \text{ if } y[k] < 0 \end{cases}$$

Without noise and ISI, the received bit sequence $\hat{b}[k]$ is thus identical to transmitted bit sequence $\hat{b}[k]$.

Although the rectangular pulse shape is a simple and intuitive choice for the transmit pulse, the significant sidelobes of the corresponding si function in the frequency domain are a major drawback:

This can be seen in the plot of the spectrum $R(f)$ of the received signal $r(t)$. The sidelobes create significant out-of-band emissions that could interfere with signals

transmitted in neighboring frequency bands. Therefore, the rectangular pulse shape is normally only used for military communication systems, like GPS, but almost never used in commercial communication systems.

Now gradually increase the noise level by decreasing the SNR with the corresponding slider to about 10 dB.

Observe that the received signal now becomes much noisier. This can be seen in the plots of the received signal $r(t)$, the spectrum before and after the matched filter, the eye diagram, and the received soft symbols $y[k]$.

Due to the noise, the eye opening is almost closed. The received soft symbols now substantially deviate from $\pm E_p = \pm 1$. However, they seldomly cross the threshold which results in a **bit error rate (BER)** that is still lower than 10^{-3} . (Note, that in the plot window the $\log_{10}$ of the BER is shown, which should thus be smaller than -3 .) Therefore, you will hardly ever notice a bit error in the plots of the Tx and Rx bits. This shows how effective digital communication systems are, even in the presence of noise.

Note down the value shown for $\log_{10}(\text{BER})$ of the BER for an SNR of 10 dB and calculate and denote the corresponding BER value.

If you further lower the SNR, first bit errors will be visible at an SNR of about 7dB, where the BER will be around 10^{-2} . If you lower the SNR even further, the BER will increase further. At an SNR of 0dB, the eye diagram has no visible structure anymore and, in the frequency domain, the spectrum of the received signal is almost completely buried in noise.

GNUradio Simulation with **RRCS Pulse Shape**

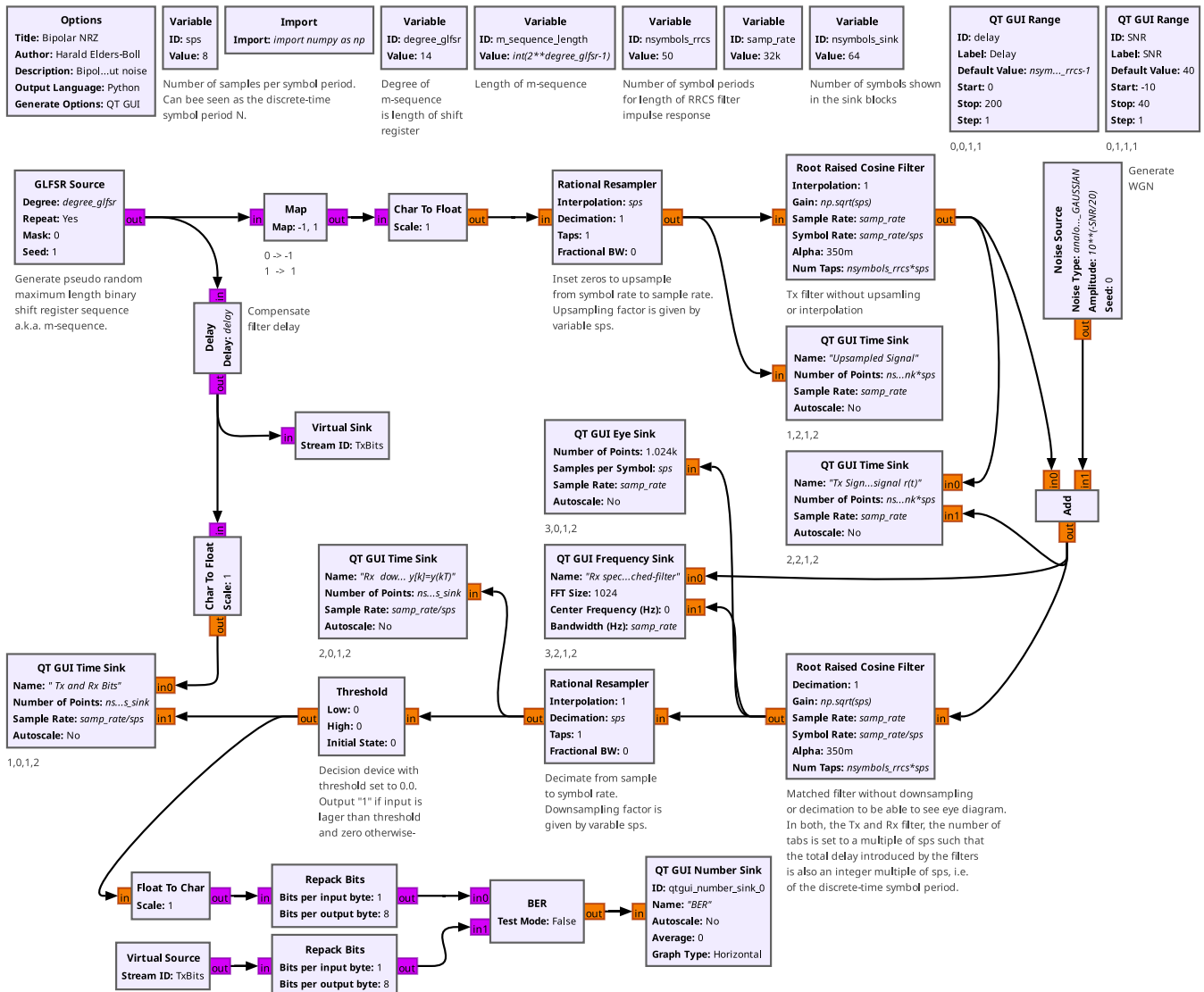
Because the rectangular transmit pulse exhibits large sidelobes in the frequency domain, it is replaced by **root-raised-cosine (RRCS)** pulse in the next GNUradio flowgraph.

The root-raised-cosine pulse shape is the most popular pulse shape in digital communication systems, because it occupies only a limited bandwidth in the frequency domain without introducing **intersymbol interference (ISI)**. The RRCS pulse does not introduce ISI since it fulfills the first **Nyquist theorem**, which requires that the autocorrelation function $\varphi_{pp}^E[n] = p[-n] * p[n]$ has equidistant zeros at multiples of the (discrete-time) symbol period N :

$$\varphi_{pp}^E[k \cdot N] = \begin{cases} E_p & , \text{ if } k = 0 \\ 0 & , \text{ if } k \neq 0 \end{cases}$$

Since the impulse response of the **root-raised cosine (RRCS)** filter is an even function, the same filters can be used for pulse shaping in the transmitter and as the matched filter in the receiver.

Now start the GNUradio companion (gnuradio-companion) and open the GNUradio flowgraph BipolarNRZ.grc :



Examine the flowgraph carefully and read the comments below the blocks to understand it well enough!

Run the flowgraph and identify the different signals that are being plotted in the window that opens up.

The RRCS transmit filter $p[n]$ and the corresponding matched-filter $h[n] = p[-n]$ both introduce a transmission delay that needs to be compensated for by the Delay block through which the transmit bits $b[k] \in \{0,1\}$ are piped before being displayed together with the received bits $\hat{b}[k]$ in the QT GUI Time Sink.

The delay introduced by the Delay block can be set by the GUI slider in the top-left corner of the plot window. If

you start the simulation, this parameter is not configured correctly to compensate for the complete transmission delay.

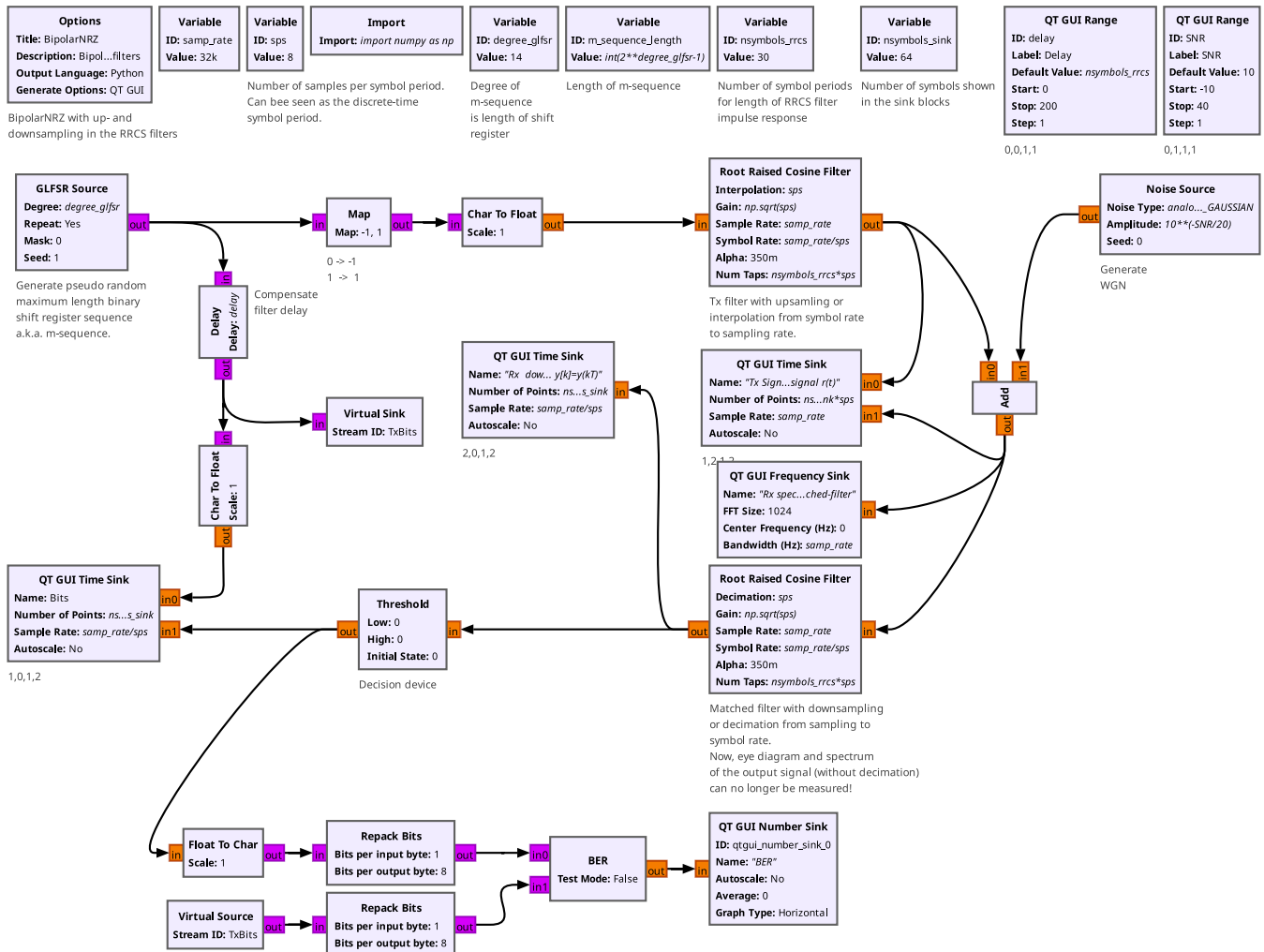
Carefully change the value of the Delay parameter step by step with the arrow key until the stem plots of transmitted and received bits (blue and red color) perfectly align. Then, you will only see one stem plot because the first plot hides the second plot.

Answer the following questions and note the answers:

Which value of the Delay parameter perfectly compensates the delay introduced by the filters? To the value of which other top-level parameter set in one of the Variable blocks in the top row of the flowgraph does this value correspond to?

GNUradio Simulation with **Resampling in the RRCS Filters**

The activation and execution of the different blocks of a flowgraph imposes an **unavoidable computational overhead because of the involved context switches**. Moreover, this scheduling is not done very efficiently in GNUradio compared to similar dataflow simulation tools. Therefore, native GNUradio blocks often implement multiple signal processing tasks. This is also true for RRCS blocks implementing the transmit and the matched filter. These can be configured to also perform the upsampling operation before the actual TX filter and the corresponding downsampling operation after the matched-filter in the receiver. This can be seen from the GNUradio flowgraph `BipolarResamplingInRRCS.grc` :



Open and run this flowgraph, to see that this results in the same signals at the other points of the flowgraph. Now, the upsampled symbol sequence $a_{up}[n]$ is no longer visible as a top-level signal, the same holds for the matched-filter output signal $y[n]$ at the sample rate r_s . Therefore, the eye diagram can no longer be displayed.

The downsampling operation takes the first of every N samples and discards the remaining $N-1$ samples. As shown by the eye diagram in the previous simulation, this corresponds to the optimum sampling point with the maximum eye-opening.

That is because the delay introduced by the filters is an integer multiple of N the number of samples per symbol. However, in a free running real-world receiver, the

optimum sampling point is not known a priori and may correspond to any of the N samples per symbol period!

Here, the value of the Delay parameter is already set to the optimum value and the initial value of the SNR is already set to 10 dB.

Note down the value show for $\log_{10}(BER)$ of the BER for the SNR of 10 dB and calculate the BER. Also note down the value for the BER for an SNR of 10 dB.

Compare this value to the BER for an SNR of 10 dB with the rectangular transmit pulse. What do you find?

Using your findings, try to **answer** the following question:

Assuming that the transmit pulse of a communication system fulfills the first Nyquist criterion, does the BER of such a communication system then depend on the pulse shape?

